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the circuit continues with its output in the high state. The output of the 555 timer goes low only when the voltage in the reset comparator exceeds $(2/3)V_{CC}$. When the output of the 555 timer is low, the capacitor cannot discharge through pin 7 since this pin is not used in this configuration. Thus, the 555 timer cannot generate additional pulses. Therefore, the only way to generate a new pulse is to turn off the circuit and then turn it on again.

Many people have attempted to create automatic dispensers using an Arduino with an infrared module, while others have used just the module itself. However, using an Arduino Nano for this purpose can be more expensive than using a 555 timer. Some dispensers do not require a microcontroller or an Arduino; they work directly. If you put

your hand near the dispenser and keep it there, the dispenser will dispense soap or gel until you remove your hand, potentially resulting in excess usage.

A commercial automatic dispenser dispenses soap or gel for a specific amount of time. Even if you trigger the sensor multiple times, the dispenser waits until you remove your hand from the sensor before allowing you to dispense again.

Circuit and working

Fig. 2 presents the circuit diagram of the automatic dispenser. It consists of an IR transmitter (emitter), an IR receiver (photodiode), rectifier diodes 1N4004 (D3 and D4), pnp transistor 2N3906 (T1), npn transistor 2N2222 (T2), an LM555 (IC1), a 5V DC motor, and a few other components.

The sensor of the dispenser is a combination of an IR emitter diode and a photodiode placed face to face, so they have line of sight, the light of IR emitter diode we cannot see with eyes. When the emitter and diode have line of sight, the photodiode conducts energy because the light of the emitter triggers the photodiode and when some object or a hand interrupts line of sight the photodiode does not conduct.

The pnp transistor 2N3906 conducts only when its base pin is at 0V or ground voltage. If it has 5V at its base, it will not conduct. Therefore, when we place our hand between the emitter and the photodiode, the base of the pnp transistor goes to 0V. Consequently, the output of the 555 timer at pin 3 generates a pulse, which is then